

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1507

Dpc Elroy 138 kV
July 21, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1507. The Interconnection Customer proposes a 113.2 MW solar photovoltaic facility interconnecting at the Dpc Elroy 138 kV substation in the Virginia area. The request was received on June 28, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 371 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,305,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Kimberley Substation - Elmont 138 kV	Bear Island - Chancellor Row 345 kV line	115.7	273	5.1
Northern Neck 345-138 kV Transformer	Chancellor Row - Skaggs Island Jct 2 345 kV	111.6	496	6.7
Humacao 2601 - Strathmore Substation 138 kV	Fritch - Lakeside 138 kV L	109.6	256	5.7
Cedar Falls Wi - Poe 138 kV	Mccartney - Yarnen 345 kV	108.2	217	7.6
Ione Energy - Poe 138 kV L	Mccartney - Yarnen 345 kV Line	106.2	205	7.6
Cogentrix - Ione Energy 138 kV L	Yarnen Substation - Mccartney 345 kV	105.1	195	7.6
Muscoda Industrial Substation - Bear Island 138 kV Line	Chncellor Row - Bear Island 345 kV L	104.4	318	5.0
Hardy Reit - Stepdown 138 kV	Mitchell Dam - Skaggs Island Jct 2 345 kV	100.8	408	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kimberley Substation - Elmont 138 kV	Rebuild line with higher-capacity conductor	\$2,586,000	3% (\$68,000)
Northern Neck 345-138 kV Transformer	Replace transformer with higher-rated unit	\$13,691,000	3% (\$395,000)
Humacao 2601 - Strathmore Substation 138 kV	Reconductor line	\$575,000	2% (\$9,000)
Poe - Cedar Falls Wi 138 kV	Reconductor line	\$2,610,000	11% (\$291,000)
Ione Energy - Poe 138 kV L	Reconductor line	\$2,095,000	11% (\$234,000)
Cogentrix - Ione Energy 138 kV L	Reconductor line	\$1,328,000	11% (\$148,000)
Muscoda Industrial - Bear Island 138 kV	Reconductor line	\$2,274,000	3% (\$60,000)
Stepdown - Hardy Reit 138 kV	Reconductor line	\$2,206,000	4% (\$100,000)
	Total allocated to project		\$1,305,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.